SLOPE CALCULATION APPLICATION V24

Installation, User and Build Manual

Platform	Build requirements
Windows .exe	Windows 10/11, Node.js LTS, npm, internet for dependencies
Android .apk	Computer, Node.js LTS, Android Studio, Android SDK
iPhone .ipa	Mac computer, Xcode and Apple signing configuration

Important: This package contains source files and build instructions. It does not contain prebuilt, signed .exe, .apk or .ipa binaries.

1. Package contents

• Slope_Calculation_Application_V24.html — main V24 app.
• Excel template — if included in the ZIP.
• Windows_EXE_Source — Electron source and Windows build script.
• Mobile_Android_iPhone_Source — Capacitor source and build instructions.
• PWA_Installable_Web_App — files for HTTPS-based browser installation.

2. Open the HTML application

1. Extract the ZIP on a computer.
2. Open Slope_Calculation_Application_V24.html in Chrome or Edge.
3. Confirm units, project information and input values before calculating.
4. Test exports and save backup copies of important project data.

3. Create Windows .exe

1. On Windows 10/11 install Node.js LTS.
2. Open Windows_EXE_Source.
3. Double-click BUILD-WINDOWS.bat.
4. Wait for npm install and Electron Builder to finish.
5. Find the installer/portable build in the dist folder.
6. Test startup, screen resizing, calculation, export and uninstall. Self-built unsigned apps may trigger Windows security warnings.

4. Create Android .apk

1. Install Node.js LTS and Android Studio/SDK.
2. Open Mobile_Android_iPhone_Source in a terminal.

3. Create www and copy index.html into it.
4. Run npm install.
5. Run npx cap add android (first time only).
6. Run npx cap sync android, then npx cap open android.
7. In Android Studio choose Build → Build Bundle(s) / APK(s) → Build APK(s).
8. Configure release signing as required and test on a real phone.

5. Create iPhone .ipa

1. Use a Mac with Xcode.
2. Create www and copy index.html into it; run npm install.
3. Run npx cap add ios (first time only), then npx cap sync ios.
4. Run npx cap open ios.
5. In Xcode configure signing, then use Product → Archive and follow Apple's export/distribution workflow.
6. Test on a real iPhone. iPhone cannot install an APK.

6. PWA quick install

Host PWA_Installable_Web_App on an HTTPS website. Android Chrome may offer Install app/Add to Home screen. On iPhone Safari, tap Share → Add to Home Screen. A PWA is not the same as a native signed APK or IPA.

7. QA checklist before field use

• Independently verify representative slope/crossfall calculations.
• Check units, rounding and invalid/blank input behavior.
• Test the sticky header and layout on small screens.
• Test every Excel/CSV/PDF export and file permission.
• Test offline behavior only if required.
• Test on physical Windows, Android and iPhone devices.
• Keep backups and follow project-approved surveying procedures.

Release status and limitations

Native binary generation and signing require platform-specific tools. This environment does not have Windows packaging tools, Android SDK or Xcode, so no prebuilt/device-tested .exe, .apk or .ipa is claimed. Successful compilation does not itself prove engineering correctness; validate calculations before project use.